Coding club recruitment 2026-27

Adaptive Fitness and Nutrition Multi-Agent System

Bonus task report **By Jahnavi Harsha**

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1. Introduction

This report documents the development of a Multi-Agent System that generates personalized daily fitness and nutrition plans. The system takes a user profile as input and produces a complete daily plan including a workout routine and meal plan. It dynamically adjusts its recommendations based on the user's goals, fatigue level, injuries and dietary restrictions.

The system was built using Groq API with the Llama 3.3 language model, ExerciseDB API for real exercise data, USDA FoodData Central API for real nutritional data and Tavily Search API for web research when injuries are detected.

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2. System Architecture

The system consists of three agents that work together.

Orchestrator Agent

This is the central coordinator. It receives the user profile, checks for injuries, delegates tasks to the Coach and Dietitian agents and combines their outputs into one complete daily plan. It is the only agent that communicates directly with the user.

Coach Agent

This agent acts as a personal trainer. It receives the user profile from the Orchestrator, fetches relevant exercises from the ExerciseDB API based on the user's goal and body part focus, and generates a customized workout plan. The plan adjusts intensity based on the user's fatigue level.

Dietitian Agent

This agent acts as a nutritionist. It receives the user profile from the Orchestrator, fetches nutritional data from the USDA FoodData Central API for relevant foods, and generates a meal plan with approximate calories that aligns with the user's goal and dietary restrictions.

The workflow is as follows. The user provides their profile. The Orchestrator receives it and checks for injuries. If an injury is found, Tavily searches the web for safe exercise modifications. The Orchestrator then calls the Coach and Dietitian agents in parallel. Both agents fetch real data from their respective APIs and generate their plans using the Llama 3.3 language model. The Orchestrator combines both outputs into a final cohesive daily plan and returns it to the user.

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3. Tools and APIs Used

ExerciseDB API via RapidAPI

Provides access to over 1,300 exercises with information about target muscle groups and body parts. The Coach Agent uses this to fetch exercises relevant to the user's goal. For example, if the goal is muscle gain, exercises targeting the chest and upper body are fetched. If a shoulder injury is present, exercises targeting upper arms are fetched instead.

USDA FoodData Central API

Provides verified nutritional data for thousands of food items including calories, protein, carbohydrates and fat content. The Dietitian Agent uses this to reference real nutritional values when creating meal plans. This ensures the calorie and macronutrient information in the plan is grounded in real data rather than generated estimates.

Tavily Search API

Provides real-time web search capability. The Orchestrator uses this when an injury is detected in the user profile. It searches for safe exercise modifications specific to that injury and passes the results to the Coach Agent. This is the multi-hop tool use component of the system. The first tool call detects the injury, the Tavily search returns safe modification guidelines, and the Coach Agent uses those guidelines to build an injury-aware workout plan.

Groq API with Llama 3.3

Powers all three agents for natural language plan generation. Groq was chosen because it provides free access with generous rate limits, making it practical for development and testing.

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4. Adaptive Features

Injury Adaptation

When the user profile contains an injury, the Orchestrator triggers a Tavily web search before passing the profile to the Coach Agent. The search results contain real information about safe exercises and modifications for that specific injury. This information is included in the Coach Agent's prompt so the generated workout plan avoids harmful movements.

Fatigue Adaptation

The user provides a fatigue level on a scale of 1 to 10. The Coach Agent's prompt includes explicit instructions to reduce workout intensity when fatigue is above 7. A user with fatigue level 9 will receive a lighter workout with fewer sets and lower intensity than a user with fatigue level 3 even if both have the same goal.

Dietary Adaptation

The Dietitian Agent adjusts meal recommendations based on the user's dietary restrictions. A vegetarian user will not receive meal plans containing meat. The dietary restriction field is included directly in the Dietitian Agent's prompt.

Goal Adaptation

Both agents adapt to the user's stated goal. A user with a muscle gain goal receives high protein meal suggestions and strength focused workouts. A user with a weight loss goal receives calorie controlled meal suggestions and cardio focused workouts.

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5. Sample Test Results

Test 1 — Normal User

Input: 22 year old male, 70kg, goal is muscle gain, beginner fitness level, no injuries, fatigue level 4, vegetarian diet.

Output: The system generated a chest and upper body workout with specific sets and reps appropriate for a beginner. The meal plan included high protein vegetarian options like paneer, lentils and eggs with approximate calorie counts for each meal.

Test 2 — Injury Adaptation

Input: 25 year old female, 60kg, goal is weight loss, beginner fitness level, shoulder injury, fatigue level 7, no dietary restrictions.

Output: The Orchestrator detected the shoulder injury and triggered a Tavily web search. The system correctly avoided shoulder exercises and recommended lower body and cardio focused movements instead. The high fatigue level of 7 resulted in a lighter intensity plan. The meal plan was adjusted for a calorie deficit appropriate for weight loss.

The difference between Test 1 and Test 2 confirms that the system successfully adapts its recommendations based on changing user constraints.

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6. GitHub Repository

The complete code, requirements file and setup instructions are available at the GitHub repository submitted alongside this report. The repository contains the bonus notebook with all agent code, the requirements.txt file listing all dependencies and a README with setup instructions.

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7. Conclusion

The Multi-Agent System successfully generates personalized fitness and nutrition plans that adapt to user injuries, fatigue levels, goals and dietary restrictions. The use of real API data from ExerciseDB and USDA ensures the plans are grounded in verified information rather than purely generated content. The multi-hop tool use through Tavily demonstrates the system's ability to handle constraints outside its core knowledge by searching the web for contextual health information.